Arpit Gupta

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Research Interests

My research aims to make a secure, performant, and affordable Internet accessible to everyone. Since my Ph.D., my group has pursued this mission along two paths—each of which requires bridging the fundamental data gaps in how we measure and operate networks:

Data-Driven Policymaking Building the measurement systems and datasets that help policymakers direct limited public resources to underserved communities (BQT; the $10B Connect America Fund audit).

Self-Driving Networks Developing production-ready ML that lets networks operate reliably at low operational cost (netUnicorn, Trustee, network foundation models).

Ph.D. work with N. Feamster and J. Rexford: software-defined Internet exchange points (SDX/iSDX) and query-driven streaming telemetry (Sonata).

Education

May'13-Aug'18 **Princeton University**, *Ph.D.*, Computer Science
Dissertation: Flexible and Scalable Systems for Network Management
Adviser: Nick Feamster
Honorable Mention, SIGCOMM Doctoral Dissertation Award

Jan'15-Jun'16 **Princeton University**, *M.A.*, Computer Science

Aug'10-May'13 **NC State University**, *M.S.*, Computer Science

Aug'05-May'09 **Indian Institute of Technology, Roorkee**, *B.Tech.*, Electronics & Comm

Professional Experience

Jul'25– **University of California, Santa Barbara**, *Associate Professor*, Santa Barbara, CA

Jul'19–Jun'25 **University of California, Santa Barbara**, *Assistant Professor*, Santa Barbara, CA

Apr'24– **Lawrence Berkeley National Lab**, *Faculty Scientist*, Berkeley, CA

Apr'25– **Benton Institute for Broadband & Society**, *Marjorie & Charles Benton Opportunity Fund Fellow*, Washington, DC

Oct'22– **Beegol Inc.**, *Member, Technical Advisory Board*, Olímpia, Sao Paulo, Brazil

Sep'18–Aug'19 **Columbia University**, *Postdoctoral Research Scientist*, New York, NY

Jan'15–Aug'18 **Princeton University**, *Graduate Research/Teaching Assistant*, Princeton, NJ

May'16–Aug'16 **Microsoft Research**, *Research Intern*, Redmond, WA

May'13–Dec'14 **Georgia Tech**, *Graduate Research/Teaching Assistant*, Atlanta, GA

Aug'11–May'12 **NC State University**, *Graduate Research/Teaching Assistant*, Raleigh, NC

May'11-Aug'11 **Google**, *Software Engineering Intern*, Mountain View, CA

Feb'10-Aug'10 **Indian Institute of Science**, *Project Assistant*, Bangalore, India

Awards & Recognitions

Students

2025 **Department of Energy GEM Fellowship**—Laasya Koduru

2024 **SIGCOMM Doctoral Dissertation Award**—Udit Paul
2024 **NSF Graduate Research Fellowship (GRFP)**—Haarika Manda
2023 UCSB CS Outstanding Dissertation Award—Udit Paul
2023 UCSB CS Outstanding Publication Award—Udit Paul
2023 UCSB CS Summer Fellowship—Roman Beltiukov
2022 Google's M-Lab Research Fellowship—Sanjay Chandrasekaran
2022 UCSB CS Outstanding Graduate Research Award—Sanjay Chandrasekaran
2022 UCSB CS Outstanding Undergraduate Research Award—Chaofan Shou
2022 UCSB CS Outstanding Teaching Assistant Award—Rohan Bhatia

Self

2026 **UC Presidential Faculty Fellows Program (PFFP)**
2025 **NSF CAREER Award**
2025 **Google ML and Systems Junior Faculty Award**
2025 **Google Research Scholar Award**
2025 **Marjorie & Charles Benton Opportunity Fund Fellowship**
2025 **IETF/IRTF Applied Networking Prize**–CAF Study
2023 **IETF/IRTF Applied Networking Prize**—TrusteeML
2022 **ACM SIGCOMM IMC Distinguished Paper Award (Long)**
2022 Honorable Mention, ACM CCS Best Paper
2018 Honorable Mention, SIGCOMM Dissertation Award
2017 Best Paper Award winner, ACM SOSR
2016 USENIX "Best of the Rest" Paper Award for Best Paper in all USENIX Conferences
2016 USENIX Community Contribution Award, USENIX NSDI
2013 Internet-2 Innovation Award
2010 College of Engineering Fellowship, North Carolina State University

Public Engagement & Impact

My broadband research has informed decisions across $100B+ in U.S. federal broadband programs—adopted in the FCC's BEAD challenge process, cited in a U.S. Supreme Court amicus brief, and used by the California Public Advocates Office (whose analysis found $1B+/yr in potential consumer savings).

Policy Briefs & Reports

[1] Hernan Galperin, Francois Bar, Arpit Gupta, Elizabeth Belding, and Laasya Koduru. Broadband Affordability and the BEAD Program: Analysis and Policy Recommendations. MEDIA Project Phase 3 Report, USC Annenberg School, 2026.

[2] Arpit Gupta. Create Independent Broadband Data for Public Accountability. Policy Brief, UC Presidential Faculty Fellows Program, 2026. Presented at Congressional Staff Briefing "Building Trust in Public Institutions", Dirksen Senate Office Building, Washington DC, May 21, 2026.

[3] Kira Allmann, Elizabeth Belding, Arpit Gupta, Laasya Koduru, Tejas Narechania, and A. Nguyen. Beyond Access: Broadband Affordability & Adoption. Research Study, Virginia Joint Commission on Technology and Science (JCOTS), Virginia General Assembly, 2025.

Amicus Briefs

[4] Elizabeth Belding, Arpit Gupta, and Tejas Narechania. Brief of Amici Curiae in Support of Respondents, Wisconsin Bell v. U.S. ex rel. Heath (No. 23-1127). Amicus Brief, Supreme Court of the United States, 2024.

Op-Eds & Public Commentary

[5] Jaber Daneshamooz, Eugene Vuong, Laasya Koduru, Sanjay Chandrasekaran, and Arpit Gupta. NetGent: Agent-based Automation of Network Application Workflows. Guest post, APNIC Blog. `https://blog.apnic.net/2026/02/05/netgent-agent-based-automation-of-network-application-workflows/`, 2026.

[6] Arpit Gupta. Computing Is a Generative Discipline. Personal blog. `https://sites.cs.ucsb.edu/~arpitgupta/blogs/computing-is-a-generative-discipline.html`, 2026.

[7] Arpit Gupta. Systems for Agents, Agents for Systems. Personal blog. `https://sites.cs.ucsb.edu/~arpitgupta/blogs/systems-for-agents-agents-for-systems.html`, 2026.

[8] Arpit Gupta. What We Can't See, We Can't Fix: The Journey to Build Independent and Accessible Broadband Data. Benton Institute for Broadband & Society, 2026.

[9] Haarika Manda, Varshika Srinivasavaradhan, Laasya Koduru, Kevin Zhang, Xuanhe Zhou, Udit Paul, Elizabeth Belding, Arpit Gupta, and Tejas Narechania. Measuring Broadband Policy Success. In *Harvard Law Review Blog*, 2024.

[10] Eric Ruzomberka, David J. Love, Christopher G. Brinton, Arpit Gupta, Chih-Chun Wang, and H. Vincent Poor. Challenges and Opportunities for Beyond-5G Wireless Security. *IEEE Security & Privacy*, 2023.

Data Contributions

[11] California Public Advocates Office (Cal Advocates), CPUC. Broadband Competition and Pricing Strategies in California's Urban Markets. California Public Advocates Office, CPUC, 2026. Competition and pricing analysis conducted using UCSB's Broadband Query Tool (BQT) pricing data.

[12] Leon Yin and Aaron Sankin. Dollars to Megabits: How We Uncovered Disparities in Internet Deals. The Markup. `https://themarkup.org/show-your-work/2022/10/19/how-we-uncovered-disparities-in-internet-deals`, 2022.

Community & Policy Engagement

2024 BEAD Challenge Process—provided evidence for Merit Network Inc. to challenge the FCC's National Broadband Map for underserved regions in Michigan

2024 Motion to empower CHRED to act against digital discrimination, City of Los Angeles—our data helped catalyze the motion

2023– #OaklandUndivided—used the BQT tool to identify underserved addresses and bridge data gaps for policymaking in Oakland, CA

2023– Institute for Local Self-Reliance (ILSR)—gathered BQT data to strengthen the case for community networks

2023– Affordable broadband for multi-dwelling units—assessing broadband offerings in MDUs using the BQT tool

2023 State of broadband affordability in Santa Barbara County—report shared with the county

Selected Media Coverage

2025 "Google honors UCSB computer scientist for pioneering low-cost AI network models"—*The UCSB Current* (also UCSB College of Engineering; *Santa Barbara Independent*)

2025 "A CAREER Award to Democratize Advanced Network Management Technology"—UCSB College of Engineering & Computer Science

2024 "Federal broadband subsidies boosted rural internet, but service faded once funding ended"—*The UCSB Current* (also UCSB College of Engineering)

2022 "Paper Coauthored by Arpit Gupta Awarded Best Paper Honorable Mention at ACM CCS"—UCSB Computer Science

Publications

50+ peer-reviewed papers at highly selective venues (typical acceptance rates 15–25%), including ACM SIGCOMM, USENIX NSDI, ACM IMC, ACM CCS, NeurIPS, PVLDB, and IEEE INFOCOM/S&P; see Google Scholar for citation metrics.

Selected Publications (independent, post-Ph.D.)

2025 *Demystifying Network Foundation Models*, NeurIPS (Datasets & Benchmarks)—the community's first systematic framework for evaluating network foundation models (IEF).

2024 *The Efficacy of the Connect America Fund in Addressing US Internet Access Inequities*, ACM SIGCOMM—**IETF/IRTF ANRP'25**; audited the $10B fund across 500,000+ addresses; grounded a U.S. Supreme Court amicus brief.

2023 *Decoding the Divide: Analyzing Disparities in Broadband Plans Offered by Major US ISPs*, ACM SIGCOMM—first address-level broadband pricing dataset (BQT).

2023 *In Search of netUnicorn: A Data-Collection Platform to Develop Generalizable ML Models*, ACM CCS—widely adopted reproducibility substrate.

2022 *AI/ML for Network Security: The Emperor has no Clothes* (Trustee), ACM CCS—**Best Paper Honorable Mention**; **IETF/IRTF ANRP'23**.

2022 *The Importance of Contextualization of Crowdsourced Active Speed Test Measurements*, ACM IMC—**Distinguished Paper Award**; adopted in the FCC's BEAD challenge process.

Preprints

[13] Jaber Daneshamooz, Satyandra Guthula, Jessica Nguyen, William Chen, Sanjay Chandrasekaran, Ankit Gupta, Arpit Gupta, and Walter Willinger. NetForge: A Programmable Substrate for Bottleneck-Centric Network Data Generation, 2025. arXiv:2507.13476.

[14] Satyandra Guthula, Jaber Daneshamooz, Charles Fleming, Kesheng Wu, Walter Willinger, and Arpit Gupta. NetBurst: Event-Centric Forecasting of Bursty, Intermittent Time Series, 2025. arXiv:2510.22397.

[15] Sylee Beltiukov, Satyandra Guthula, Haarika Manda, Jaber Daneshamooz, Wenbo Guo, Walter Willinger, Arpit Gupta, and Inder Monga. netFound: Principled Design for Network Foundation Models, 2023. arXiv:2310.17025 (latest revision v5, 2026).

[16] Hyeongseong Lee, Udit Paul, Arpit Gupta, Elizabeth Belding, and Mengyang Gu. Analyzing Disparity and Temporal Progression of Internet Quality through Crowdsourced Measurements with Bias-Correction, 2023. arXiv:2310.16136.

[17] Rohan Bhatia, Arpit Gupta, Rob Harrison, Daniel Lokshtanov, and Walter Willinger. DynamiQ: Planning for Dynamics in Network Streaming Analytics Systems, 2021. arXiv:2106.05420.

[18] Udit Paul, Jiamo Liu, Vivek Adarsh, Mengyang Gu, Arpit Gupta, and Elizabeth Belding. Characterizing Performance Inequity Across U.S. Ookla Speedtest Users, 2021. arXiv:2110.12038.

Conference Papers

[19] Haarika Manda, Manshi Sagar, Yogesh, Kartikay Singh, Cindy Zhao, Tarun Mangla, Phillipa Gill, Elizabeth Belding, and Arpit Gupta. TURBOTEST: Learning When Less is Enough through Early Termination of Internet Speed Tests. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2026.

[20] Murayyiam Parvez, Annus Zulfiqar, Roman Beltiukov, Shir Landau Feibish, Walter Willinger, Arpit Gupta, and Muhammad Shahbaz. SpliDT: Partitioned Decision Trees for Scalable Stateful Inference at Line Rate. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2026.

[21] Sylee Beltiukov, Satyandra Guthula, Wenbo Guo, Walter Willinger, and Arpit Gupta. Demystifying Network Foundation Models. In *Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. arXiv:2509.23089.

[22] Laasya Koduru, Arpit Gupta, Elizabeth Belding, and Tejas Narechania. Evaluating the Effects of and Interdependencies among Federal Broadband Funding Programs. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, 2025.

[23] Laasya Koduru, A. Rojas, A. Penate, Z. Zhou, Francois Bar, Elizabeth Belding, Hernan Galperin, and Arpit Gupta. Assessing the Broadband Service Gaps and Affordability Barriers in BEAD-Eligible Areas. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, 2025.

[24] Congcong Miao, Zhizhen Zhong, Yiren Zhao, Arpit Gupta, Ying Zhang, Sirui Li, Zekun He, Xianneng Zou, and Jilong Wang. PreTE: Traffic Engineering with Predictive Failures. In *ACM SIGCOMM*, 2025.

[25] Murayyiam Parvez, Annus Zulfiqar, Roman Beltiukov, Shir Landau Feibish, Walter Willinger, Arpit Gupta, and Muhammad Shahbaz. SpliDT: Partitioned Decision Trees for Scalable Stateful Inference at Line Rate. In *ACM SIGCOMM (Short Papers)*, 2025.

[26] Z. Wen, J. Bliton, Laasya Koduru, Elizabeth Belding, Tejas Narechania, Arpit Gupta, and Shaddi Hasan. Strategic Misreporting in the National Broadband Map. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, 2025.

[27] Walter Willinger, Ronaldo A. Ferreira, Arpit Gupta, Roman (Sylee) Beltiukov, Satyandra Guthula, Lisandro Z. Granville, and Arthur S. Jacobs. When something looks too good to be true, it usually is! ai is causing a credibility crisis in networking. *ACM SIGCOMM Computer Communication Review (CCR)*, 2025.

[28] Momin Haider, Ming Yin, Menglei Zhang, Arpit Gupta, Jing Zhu, and Yu-Xiang Wang. Networkgym: Reinforcement learning environments for multi-access traffic management in network simulation. In *The Thirty-eight Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.

[29] Jiamo Liu, David Lerner, Jae Chung, Udit Paul, Arpit Gupta, and Elizabeth Belding. Watching Stars in Pixels: The Interplay of Traffic Shaping and YouTube Streaming QoE over GEO Satellite Networks. In *International Conference on Passive and Active Network Measurement*, pages 153–169. Springer, 2024.

[30] Haarika Manda, Varshika Srinivasavaradhan, Laasya Koduru, Kevin Zhang, Xuanhe Zhou, Udit Paul, Elizabeth Belding, Arpit Gupta, and Tejas Narechania. The Efficacy of the Connect America Fund in Addressing US Internet Access Inequities. In *ACM SIGCOMM*, 2024.
Building block for "Amici Curiae Brief in Support of Respondents, Wisconsin Bell v. U.S. ex rel. Heath (No. 23-1127) (U.S. Oct. 1, 2024)"
IETF/IRTF'25 Applied Networking Research Prize (ANRP) .

[31] Chris Misa, Ram Durairajan, Arpit Gupta, Reza Rejaie, and Walter Willinger. Leveraging Prefix Structure to Detect Volumetric DDoS Attack Signatures with Programmable Switches. In *IEEE Symposium on Security and Privacy (S&P)*, 2024.

[32] Chaofan Shou, Rohan Bhatia, Arpit Gupta, Rob Harrison, Daniel Lokshtanov, and Walter Willinger. Query planning for robust and scalable hybrid network telemetry systems. In *ACM Conference on Emerging Networking Experiments and Technologies (CoNEXT)*, 2024.

[33] Roman Beltiukov, Wenbo Guo, Arpit Gupta, and Walter Willinger. In Search of netUnicorn: A Data-Collection Platform to Develop Generalizable ML Models for Network Security Problems. In *ACM Conference on Computer and Communications Security (CCS)*, 2023.

[34] Udit Paul, Vinothini Gunasekaran, Jiamo Liu, Tejas Narechania, Arpit Gupta, and Elizabeth Belding. Decoding the Divide: Analyzing Disparities in Broadband Plans Offered by Major US ISPs. In *ACM SIGCOMM*, 2023.
Part of Udit Paul's Dissertation—ACM SIGCOMM Dissertation Award.

[35] Taveesh Sharma, Tarun Mangla, Arpit Gupta, Junchen Jiang, and Nick Feamster. Estimating webrtc video qoe metrics without using application headers. In *ACM SIGCOMM Internet Measurement Conference (IMC)*, 2023.

[36] Fuheng Zhao, Punnal Ismail Khan, Divyakant Agrawal, Amr El Abbadi, Arpit Gupta, and Zaoxing Liu. Panakos: Chasing the tails for multidimensional data streams. In *International Conference on Very Large Data Bases (VLDB)*, 2023.

[37] Arthur S Jacobs, Roman Beltiukov, Walter Willinger, Ronaldo A Ferreira, Arpit Gupta, and Lisandro Z Granville. Ai/ml for network security: The emperor has no clothes. In *ACM Conference on Computer and Communications Security (CCS)*, 2022.
IETF/IRTF'23 Applied Networking Research Prize (ANRP)
ACM CCS'22 Best Paper Award (Honorary Mention).

[38] Congcong Miao, Minggang Chen, Arpit Gupta, Zili Meng, Lianjin Ye, Jingyu Xiao, Jie Chen, Zekun He, Xulong Luo, Jilong Wang, and Heng Yu. Detecting Ephemeral Optical Events with OpTel. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2022.

[39] Udit Paul, Jiamo Liu, David Farias-Llerenas, Vivek Adarsh, Arpit Gupta, and Elizabeth Belding. Characterizing Internet Access and Quality Inequities in California M-Lab Measurements. In *ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS)*, 2022.

[40] Udit Paul, Jiamo Liu, Arpit Gupta, Mengyang Gu, and Elizabeth Belding. The Importance of Contextualization of Crowdsourced Active Speed Test Measurements. In *ACM SIGCOMM Internet Measurement Conference (IMC)*, 2022.
ACM SIGCOMM IMC'22 Distinguished Paper Award
Part of Udit Paul's Dissertation—ACM SIGCOMM Dissertation Award.

[41] Vivek Adarsh, Michael Nekrasov, Udit Paul, Tarun Mangla, Arpit Gupta, Morgan Vigil-Hayes, Ellen W Zegura, and Elizabeth M Belding. Coverage is Not Binary: Quantifying Mobile Broadband Quality in Urban, Rural and Tribal Contexts. In *International Conference on Computer Communications and Networks (ICCCN)*, 2021.

[42] Todd Arnold, Ege Gürmeriçliler, Arpit Gupta, Matt Calder, Georgia Essig, Vasileios Giotsas, and Ethan Katz-Bassett. (How Much) Does a Private WAN Improve Cloud Performance? In *IEEE Conference on Computer Communications (INFOCOM)*, 2020.

[43] Rob Harrison, Shir Landau Feibish, Arpit Gupta, Ross Teixeira, S. Muthukrishnan, and Jennifer Rexford. Carpe Elephants: Seize the Global Heavy Hitters. In *ACM SIGCOMM Workshop on Secure Programmable Network Infrastructure (SPIN)*, 2020.

[44] Ross Teixeira, Rob Harrison, Arpit Gupta, and Jennifer Rexford. PacketScope: Monitoring the Packet Lifecycle Inside a Switch. In *ACM Symposium on SDN Research (SOSR)*, 2020.

[45] Arpit Gupta, Rob Harrison, Ankita Pawar, Marco Canini, Nick Feamster, Jennifer Rexford, and Walter Willinger. Sonata: Query-Driven Streaming Network Telemetry. In *ACM SIGCOMM*, 2018.

[46] Xiaohe Hu, Arpit Gupta, Nick Feamster, Aurojit Panda, and Scott Shenker. Preserving Privacy at IXPs. In *ACM APNet*, 2018.

[47] Robert MacDavid, Rüdiger Birkner, Ori Rottenstreich, Arpit Gupta, Nick Feamster, and Jennifer Rexford. Concise Encoding of Flow Attributes in SDN Switches. In *ACM Symposium on SDN Research (SOSR)*, 2017.
ACM SOSR'17 Best Paper Award.

[48] Arpit Gupta, Robert MacDavid, Rüdiger Birkner, Marco Canini, Nick Feamster, Jennifer Rexford, and Laurent Vanbever. An Industrial-Scale Software Defined Internet Exchange Point. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2016.
NSDI'16 Community Award
USENIX'16 Best of the Rest.

[49] Hyojoon Kim, Joshua Reich, Arpit Gupta, Muhammad Shahbaz, Nick Feamster, and Russ Clark. Kinetic: Verifiable Dynamic Network Control. In *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2015.

[50] Arpit Gupta, Laurent Vanbever, Muhammad Shahbaz, Sean Patrick Donovan, Brandon Schlinker, Nick Feamster, Jennifer Rexford, Scott Shenker, Russ Clark, and Ethan Katz-Bassett. SDX: A Software Defined Internet Exchange. In *ACM SIGCOMM*, 2014.
Internet2 Innovation Award.

[51] Arpit Gupta, Jeongki Min, and Injong Rhee. Wifox: Scaling wifi performance for large audience environments. In *ACM Conference on Emerging Networking Experiments and Technologies (CoNEXT)*, 2012.

Workshop & Short Papers

[52] Jaber Daneshamooz, Eugene Vuong, Laasya Koduru, Sanjay Chandrasekaran, and Arpit Gupta. NetGent: Agent-Based Automation of Network Application Workflows. In *NeurIPS Workshop on Machine Learning for Systems (MLforSys)*, 2025.

[53] Punnal Ismail Khan, Satyandra Guthula, Roman Beltiukov, Roland Schmid, Tobias Bühler, Arpit Gupta, Laurent Vanbever, and Walter Willinger. Harnessing public code

repositories to develop production-ready ml artifacts for networking. In *Proceedings of the Applied Networking Research Workshop*, 2024.

[54] Roman Beltiukov, Sanjay Chandrasekaran, Arpit Gupta, and Walter Willinger. Pinot: Programmable infrastructure for networking. In *Proceedings of the Applied Networking Research Workshop*, 2023.

[55] Arpit Gupta, Ramakrishnan Durairajan, and Walter Willinger. Special issue on the acm sigmetrics workshop on measurements for self-driving networks. *ACM SIGMETRICS Performance Evaluation Review*, 2023.

[56] Walter Willinger, Arpit Gupta, Roman Beltiukov, and Wenbo Guo. A NetAI Manifesto (Part II): Less Hubris, More Humility. In *ACM SIGMETRICS Workshop on Measurements for Self-Driving Networks*, 2023.

[57] Walter Willinger, Arpit Gupta, Arthur S. Jacobs, Roman Beltiukov, Ronaldo A. Ferreira, and Lisandro Z. Granville. A NetAI Manifesto (Part I): Less Explorimentation, More Science. In *ACM SIGMETRICS Workshop on Measurements for Self-Driving Networks*, 2023.

[58] Palak Goenka, Kyriakos Zarifis, Arpit Gupta, and Matt Calder. Towards client-side active measurements without application control. *ACM SIGCOMM Computer Communication Review (CCR)*, 2022.

[59] Tarun Mangla, Udit Paul, Arpit Gupta, Nicole P Marwell, and Nick Feamster. Internet inequity in chicago: Adoption, affordability, and availability. *Affordability, and Availability (August 5, 2022)*, 2022.
Part of Udit Paul's Dissertation—ACM SIGCOMM Dissertation Award.

[60] Vivek Adarsh, Michael Nekrasov, Udit Paul, Alexander Ermakov, Arpit Gupta, Morgan Vigil-Hayes, Ellen W Zegura, and Elizabeth M Belding. Too Late for Playback: Estimation of Video Stream Quality in Rural and Urban Contexts. In *Passive and Active Measurement Conference (PAM)*, 2021.

[61] Zongyi Zhao, Xingang Shi, Arpit Gupta, Qing Li, Zhiliang Wang, Bin Xiong, and Xia Yin. Continuous Flow Measurement with SuperFlow. In *IEEE/ACM International Symposium on Quality of Service (IWQoS)*, 2021.

[62] Todd Arnold, Matt Calder, Italo Cunha, Arpit Gupta, Harsha V. Madhyastha, Michael Schapira, and Ethan Katz-Bassett. Beating BGP is Harder than We Thought. In *ACM HotNets*, 2019.

[63] Arpit Gupta, Chris MacStoker, and Walter Willinger. An Effort to Democratize Networking Research in the Era of AI/ML. In *ACM HotNets*, 2019.

[64] Hyojoon Kim and Arpit Gupta. Ontas: Flexible and scalable online network traffic anonymization system. In *ACM SIGCOMM NetAI*, 2019.

[65] Rob Harrison, Qizhe Cai, Arpit Gupta, and Jennifer Rexford. Network-Wide Heavy Hitter Detection with Commodity Switches. In *ACM Symposium on SDN Research (SOSR)*, 2018.

[66] Rüdiger Birkner, Arpit Gupta, Nick Feamster, and Laurent Vanbever. SDX-Based Flexibility or Internet Correctness?: Pick Two! In *ACM Symposium on SDN Research (SOSR)*, 2017.

[67] Arpit Gupta, Rüdiger Birkner, Marco Canini, Nick Feamster, Chris Mac-Stoker, and Walter Willinger. Network Monitoring as a Streaming Analytics Problem. In *ACM Workshop on Hot Topics in Networks (HotNets)*, 2016.

[68] Arpit Gupta, Nick Feamster, and Laurent Vanbever. Authorizing Network Control at Software Defined Internet Exchange Points. In *ACM Symposium on SDN Research (SOSR)*, 2016.

[69] Arpit Gupta, Matt Calder, Nick Feamster, Marshini Chetty, Enrico Calandro, and Ethan Katz-Bassett. Peering at the Internet's Frontier: A First Look at ISP Interconnectivity in Africa. In *Passive and Active Network Measurement (PAM)*, 2014.

Research Funding

Total: ∼\$5.9M in awarded research funding; ∼\$2.8M as sole/lead PI.

Making Agentic AI Safe for DOE User Facilities
Sponsor: DOE / LBNL Laboratory Directed Research & Development (LDRD)
Investigator(s): J. Wu (PI, ESnet/LBNL), A. Gupta (Co-PI)
Amount: \$ 250k for 1 year (FY27).
Awarded: 2026 (project start October 1, 2026).

Bridging the Representation–Semantics Gap for Production-Ready AI-Powered Network Operations
Sponsor: Cisco Research
Investigator(s): A. Gupta
Amount: \$ 75k.
Awarded: Fall 2025.

ML and Systems Junior Faculty Award: Low Infrastructure ML
Sponsor: Google
Investigator(s): A. Gupta
Amount: \$ 100k for 1 year.
Awarded: Summer 2025.

Research Scholar Award: Network Foundation Model for Enabling AI-powered Network Operations (AIOps)
Sponsor: Google
Investigator(s): A. Gupta
Amount: \$ 60k for 1 year.
Awarded: Summer 2025.

Characterizing Broadband Pricing in California
Sponsor: California Public Utility Commission (CPUC)
Investigator(s): A. Gupta, E. Belding
Amount: \$ 125k for 2 years.
Awarded: Summer 2025.

Characterizing Barriers to Digital Inclusion in Virginia
Sponsor: Virginia Joint Commission on Technology and Sciences (JCOTS)
Investigator(s): A. Gupta, E. Belding, and T. Narechania
Amount: \$ 30k for .5 years.
Awarded: Summer 2025.

CAREER: Developing Generalizable ML Models for Diverse Learning Problems in Network Operations
Sponsor: NSF
Investigator(s): A. Gupta
Amount: \$ 700k for 5 years.
Awarded: Spring 2025.

Marjorie & Charles Benton Opportunity Fund Fellowship: Examining Barriers to Broadband
Sponsor: Benton Institute for Broadband & Society
Investigator(s): A. Gupta
Amount: \$ 20k for 2 years.
Awarded: Spring 2025.

Measuring the Effectiveness of Digital Inclusion Approaches
Sponsor: Pew Charitable Trust
Investigator(s): A. Gupta, E. Belding
Amount: $ 42k for 1 year.
Awarded: Spring 2025.

Telemetry-driven Foundation Models for Self-Driving Networks
Sponsor: Cisco Research
Investigator(s): A. Gupta
Amount: $ 90k for 1 year.
Awarded: Spring 2024.

netFound: An AI/ML foundational model for Network
Sponsor: DoE
Investigator(s): A. Gupta and I. Monga (ESnet)
Amount: $100k GPU and 100k CPU hours for 1 year.
Awarded: Winter 2024.

IMR: MT: NetFlex: A Flexible Scalable & Privacy-Preserving Network Measurement Platform to Iteratively Collect Multi-modal Multi-view Network Data from Access Networks
Sponsor: NSF
Investigator(s): A. Gupta
Amount: $ 600k for 2 years.
Awarded: Fall 2023.

IMR: RI-P: Programmable Closed-loop Measurement Platform for Last-Mile Networks
Sponsor: NSF
Investigator(s): A. Gupta and N. Feamster
Amount: $ 100k for 2 years.
Awarded: Fall 2022.

IMR: MM-1A: ADDRESS: Augment, Denoise and Debias Crowdsourced Measurements for Statistical Synthesis of Internet Access Characterization
Sponsor: NSF
Investigator(s): E. Belding, A. Gupta, and M. Gu
Amount: $ 600k for 3 years.
Awarded: Fall 2022.

CC* Integration-Large: Democratizing Networking Research in the Era of AI/ML
Sponsor: NSF
Investigator(s): A. Gupta, T. Gupta, E. Belding, and N. Feamster
Amount: $ 1M for 3 years.
Awarded: Fall 2021.

CC* Integration-Large: Bringing Code to Data: A Collaborative Approach to Democratizing Internet Data Science
Sponsor: NSF
Investigator(s): R. Durairajan, A. Gupta, R. Rezai, J. Miyake, and D. Teach
Amount: $ 1M for 3 years.
Awarded: Fall 2021.

Scaling Network Security with PISA Switches
Sponsor: Verizon
Investigator(s): A. Gupta
Amount: $ 200k for 2 years.
Awarded: December 2019.

Self-Driving Broadband Networks
Sponsor: Beegol Inc.
Investigator(s): A. Gupta
Amount: $ 35k for 1 year.
Awarded: July 2020.

Workshop on Next-G Mobile Security
Sponsor: NSF
Investigator(s): A. Gupta
Amount: $ 49k for 1 year.
Awarded: July 2020.

MLWiNS: RL-based Self-driving Wireless Network Management System for QoE Optimization
Sponsor: NSF & Intel
Investigator(s): A. Gupta (PI), E. Belding, Y. Wang
Amount: $ 840k for 3 years.
Awarded: July 2020.

Software & Datasets

Software

NetBurst—event-centric forecasting foundation model for bursty network time series; *tech-transfer at ESnet within the DOE Genesis Mission portfolio.* `https://github.com/SNL-UCSB/NetBurst-essential`

netFound—network foundation model for security and telemetry; *in active tech-transfer at ESnet and Google (DOE Genesis Mission); on the DOE AI Model Card.* `https://github.com/SNL-UCSB/netFound`

IEF (Intrinsic Evaluation Framework)—the community's first systematic framework for evaluating network foundation models ("Demystifying Network Foundation Models", NeurIPS'25). `https://github.com/maybe-hello-world/demystifying-networks`

NetGent—agent-based automation of network application workflows. `https://github.com/SNL-UCSB/netgent`

NetForge / netReplica—programmable substrate for last-mile network data generation. `https://github.com/SNL-UCSB/netforge`

netUnicorn—data-collection platform for generalizable ML models for networking (ACM CCS'23); *adopted by research groups and industry as a reproducibility substrate.* `https://github.com/netunicorn/netunicorn`

PINOT—programmable measurement infrastructure for AI/ML for networking; *deployed at UCSB and in active use at LBNL/ESnet.* `https://pinot.cs.ucsb.edu`

Trustee—framework to extract decision-tree explanations from black-box ML models (ACM CCS'22 Best Paper HM; 2023 IRTF ANRP). `https://trusteeml.github.io`

Datasets

Broadband-plan Querying Tool (BQT) Dataset—the first comprehensive broadband-affordability dataset; *used by Cal Advocates, ILSR, and Merit Network for policy analysis.* `https://bqt.cs.ucsb.edu`

Invited Talks & Presentations

Keynotes

2025 *Making the Self-Driving "Net" Work: Developing Production-Ready ML Models for Self-Driving Networks*—INDIS Workshop, SC25 (Supercomputing), St. Louis, MO

2025 *Making the "Net" Work for All: From Measurement to Impact*—inaugural IMC Student Workshop, co-located with ACM IMC 2025

2024 *In Search of a Networking Unicorn: Realizing Closed-Loop ML Pipelines for Networking*—8th Keeping Networks Integrated Together (KNIT) Workshop, San Diego Supercomputer Center

Invited Talks

2026 Panelist (Automation; Data Generation panels)—Workshop on Generative AI in Networking, University of Chicago Paris Center, France

2026 *Making Networks Reliable with AI* (Panel)—Google India Networking Research Summit, Bengaluru, India

2025 *Demystifying Network Foundation Models*—Google Networking Research Summit, Sunnyvale, CA

2025 *Making the Self-Driving "Net" Work: Developing Production-Ready ML Models for Self-Driving Networks*—6GIC-CLICK Workshop on Telecom AI, Institute for Communication Systems, University of Surrey, UK

2024 *Making the "Net" Work for All: Enabling Secure, Performant, and Equitable Internet Services*—AWS Networking Research Seminar; Georgia Tech; University of Chicago; Purdue University

2023 *Democratizing the Development of Production-ready ML Artifacts for Networking*—IIT Roorkee; IIT Delhi; IIIT Delhi

2023 *In Search of a Networking Unicorn: Realizing Closed-loop ML Pipelines for Networking*—IISER Bhopal; Stanford University

2023 *Closed-loop ML for Networking*—Google Networking Research Summit (Panel), Sunnyvale, CA

2022 *AI/ML for Network Security: The Emperor has no Clothes*—ETH Zurich, Switzerland

2022 *Democratizing Networking Research in the Era of AI/ML*—Google Networking Research Summit (virtual)

Seminars & Workshop Talks

2026 *Agentic AI in Network Measurement and Infrastructure* (moderated session)—CAIDA AIMS Workshop, UC San Diego

2024 *In Search of a Networking Unicorn: Realizing Closed-Loop ML Pipeline for Networking*—SECDAI 2024

2023 *In Search of netUnicorn: A "Thin Waist" Data-Collection Platform for Generalizable ML Models*—Intel MLWiNS Monthly Tech Talk

2023 *Opening Remarks*, ACM SIGCOMM 2023 Tutorial: Closed-Loop "ML for Networks" Pipelines, New York, NY

2022–23 *RELIEF: RL-based Self-driving Wireless Network Management System for QoE Optimization*—MLWiNS Annual (2022), Mid-Year (2023), and Annual (2023) Workshops

Professional Activities

Program Committee Chair

2026 ACM SIGCOMM Student Research Competition (SRC)

2025 ACM SIGCOMM Non-Paper Session Track

2025 ACM IMC Reproducibility Track

2024 ACM SIGCOMM Workshop on Networks for AI Computing (NAIC)

2021 ACM SIGCOMM Artifact Evaluation Committee, Virtual

2020 ACM SIGCOMM Student Research Competition, Virtual

2019 ACM SIGCOMM Workshop on Network Meets AI & ML (NetAI), Beijing, CN

Organizing Committee Chair

2023 NSF Workshop on "Bridging the Divide: Answering Internet Policy Questions with Cutting-Edge Network Measurement Algorithms, Datasets, and Platforms", Berkeley, CA

2023 SIGMETRICS Workshop on “Measurements for Self-driving Networks", Orlando, FL

2020 NSF Workshop on NextG Security, Virtual

2019 NSF Workshop on “Measurements for Self-driving Networks", Princeton, NJ

Program Committee Member

2026 IFIP Networking—Workshop on Systems for AI / AI for Systems

2024-26 ACM SIGMETRICS

2023-25 ACM SIGCOMM IMC

2020-26 USENIX Networked System Design and Implementation (NSDI)

2020-21, 2026 ACM SIGCOMM

2026 Workshop on Trustworthy and Reproducible AI for Networking (co-located with ACM CoNEXT)

2026 ACM Workshop on Networks for AI Computing (NAIC, co-located with ACM CoNEXT)

2026 ACM Workshop on LEO Networking and Communication (LEO-NET)

2025 Workshop on Practical Adoption Challenges of ML for Systems (PACMI, co-located with ACM CoNEXT)

2019 ACM Conference on Emerging Networking Experiments and Technologies (CoNEXT)

2019 ACM SIGCOMM Symposium on SDN Research (SOSR)

2018 ACM SIGCOMM Workshop on Self-Driving Networks

Organizing Committee Member

2018 NSF Workshop on Self-driving Networks, Princeton

Steering Committee Member

2024– Cal-Bridge Doctoral Program

Panelist

2016 GENI Network Innovators Community Event 2016

2016 CITP Conference on Global Internet Interconnection

External Reviewer

2019 Privacy Enhancing Technologies Symposium (PETS)

2017 ACM SIGCOMM

2016 IEEE International Conference on Network Protocol (ICNP)

2014 USENIX Networked System Design and Implementation (NSDI)

Session Chair

2026 USENIX NSDI (“Self-Managing Networks")

2025 ACM IMC (Session 2B: “Passive Traffic Analysis")

2023 USENIX NSDI (“Making Systems Learn")

2022 ACM IMC (“Videoconferencing")

2021 ACM SIGCOMM (“Cellular and 5G Networks"); USENIX NSDI (“Datacenter Networking and SDNs")

2020 ACM SIGCOMM (“Telemetry: Tell Me More About My Packets")

Ph.D. & M.S. Committees

Ph.D.: Nawel Alioua, Jiamo Liu, Esther Showalter (UCSB)

M.S.: Vinothini Gunasekaran, Momin Haider (UCSB)

Department & University Service

2023-24 Public Relations & Awards Committee, CS Department, UCSB

2022-24 Graduate Admissions Committee, CS Department, UCSB

Teaching

Instructor

2020-25 **(New)** ML For Networked Systems, (CS 293N), UCSB

2020-26 Advanced Topics in Internet Computing, (CS 176C), UCSB

2021-23 **(New)** Programmable Networks, (CS 176B), UCSB

2023-24 **(New)** ML for Networks, (CS 190N), UCSB

Tutorials

2023 Closed-Loop “ML for Networks" Pipelines—ACM SIGCOMM 2023, New York, NY

Students

Current Ph.D. Students

Fall 2019 – * Sanjay Chandrasekaran—M-Lab Research Fellowship

Fall 2022 – * Satyandra Guthula

Fall 2023 – * Haarika Manda (co-advised by Prof. Belding)—NSF Graduate Research Fellowship (GRFP)

Fall 2023 – * Jaber Daneshamooz

Fall 2024 – * Laasya Koduru—Department of Energy GEM Fellowship

Fall 2025 – * Manni Moghimi

Past Ph.D. Students

Fall 2021 – May’26 Sylee (Roman) Beltiukov, UCSB, Ph.D.—*Dissertation*: Credible Machine Learning for Networking: Diagnosis, Data Collection, and Network-Native Representation Learning—ACM CCS’22 Best Paper Honorable Mention, IRTF’23 Applied Networking Research Prize, UCSB CS Summer Fellowship’23; *now at Google*

Fall 2018 – 23 Udit Paul, UCSB, Ph.D. (co-advised with Prof. Belding)—ACM SIGCOMM Dissertation Award’24, IMC’22 Distinguished Paper Award, UCSB CS Outstanding Publication Award’23, UCSB CS Outstanding Dissertation Award’23; *now at Cisco ThousandEyes (via Ookla)*

Current MS Students

2025 – * Shaunak G. Umarkhedi (NetVibe; M.S., Cornell University)

Past MS Students

Fall 2019 – 21 Abtin Bateni—*now at Google*

Fall 2020 – 22 Irene Pattarachanyakul

Fall 2020 – 22 Rohan Bhatia—*now at Google*

Fall 2021 – 23 Navya Battula—*now at OQ Point*

Fall 2022 – 24 Khalid Mihlar

Fall 2023 – 24 Laasya Koduru—*continued to Ph.D. at UCSB*

Fall 2023 – Winter 25 Zephyr (Xuanhe) Zhou

Spring 2024 – 25 Jessica Nguyen

Fall 2024 – 26 Alexander Nguyen (BS/MS; BQT)

Fall 2024 – 26 Abhishek Kumar

Fall 2025 – 26 Amrutha Kalle

Current Undergraduate Students

2025 – *	Srinandha Murugesan (BQT SaaS, ERSP)
2025 – *	Chen Hui Lin (BQT SaaS, ERSP)
2025 – *	Ivy Holiday (BQT SaaS, ERSP)
2025 – *	Daniel Chang (BQT SaaS, ERSP)
2025 – *	Red Macasa (NetVibe)
2025 – *	Sumedh Aerram (NetVibe)
2025 – *	Anit Annadi (NetVibe)
2026 – *	Srish Nigam (Pramana)
2026 – *	Jenil Rajeshkumar (NetVibe / agentic systems)
2026 – *	Sidharth Kalje (NetVibe)

Past Undergraduate Students

2025	Eugene Vuong (Cal-Bridge, CSU East Bay)
2024 – 25	Daphne Jomo (Cal-Bridge Summer Scholar), Ken Thampiratwong, Akul Singh, Evania Cheng, Karthik Bhattaram, Ziv Weissman
2023 – 24	Kevin Zhang (co-author, SIGCOMM'24); Robin Martin (Cal-Bridge)
2022 – 23	Aditi Phatak, Michael Yang, Abel Atnafu, Luis Bravo, Emily Hu
2021 – 22	Chaofan Shou (lead author, CoNEXT'24; now Ph.D. at UC Berkeley), Divyanshu Setia, Arnesh Agrawal, Hemant, Hemil Panchiwala, Anmol Khemuka, Ethan Wu, Sam Liang, Aseem Verma
2020 – 21	Jake Miller, Kaustubh Trivedi, Palak Goenka, Bhavye Jain, Ritik Kumar, Matthew Aragaw, John-Michael Kirchner, Ruchika Saswade
2019 – 20	Rohan Bhatia, Samagra Sharma

High School Students

2026 – *	Sthanav Murali (NetGent / Pramana; Monta Vista High School)
2024 – 25	William Chen (Cary Academy; co-author, NetReplica; now at MIT)
2021 – 22	Grant Withee (Laguna Blanca School; now at Caltech)